

JOSEPH OTIENO JUMA

Nairobi Kenya | +254721747991 | josephjuma24@gmail.com | Portfolio: <https://josephjuma24.github.io/>

Summary

AI researcher, trainer, and adoption consultant with 14+ years of industry experience, including 3+ years specializing in AI research, human-centered digital transformation, and technical training. Contributed to the development of **70+ AI-driven innovation concepts implemented across smartphone series in African markets**, bridging user needs with scalable product solutions. Expert in designing and delivering AI literacy programs, adoption strategies, and hands-on workshops for executives, researchers, educators and pastoral ministries. Skilled at translating complex technical insights into inclusive, culturally relevant, and actionable strategies that drive innovation, adoption, and trust across Africa’s technology ecosystem.

Core Skills

AI Training & Facilitation	AI Adoption Strategy
AI UX research (qualitative and quantitative)	Prompt engineering and context engineering
Ethical AI and data privacy advocacy	Coding skills (Python, Bash, Pandas, NumPy)
Data cleaning, analysis, and visualization	Data visualization using Tableau or Power BI
Cloud computing: AWS, Azure, or GCP	AI governance and compliance
Machine learning, deep learning, GenAI	Analytical and critical thinking
LLMs, multimodal AI, and AI agents	Creative and innovative thinking
ML frameworks like PyTorch and TensorFlow	Teamwork and collaboration skills

Technical Toolkit

AI Tools: • Microsoft Copilot • NotebookLM • Claude • Google Gemini • ChatGPT • Perplexity
Research: • Elicit • Scite.ai • SciSpace • Litwise • Julius • Consensus • Mendley • Originality.ai
Data Analytics: • Excel • Python Libraries • Jupyter Notebook • PowerBI • Looker • Alteryx • Qlik

Experience

AI Trainer & Adoption Consultant	01/2026 to date
Daystar Centre of AI, Modelling & Digital Trust (DCAMD)	Nairobi, Kenya
i. Delivering AI training programs (FAIDA & AI Research Clinics) to diverse post-graduate learners. ii. Facilitating hands-on sessions using AI tools for research, data analysis, and productivity. iii. Developing training materials, structured modules, and practical exercises. iv. Mentoring learners across multiple cohorts, improving AI adoption and confidence. v. Introducing AI-assisted workflows to enhance research efficiency and output. vi. Ensuring strong focus on human-centered AI adoption and context implementation. vii. Experienced in training executives, students, and non-technical users viii. Passionate about AI education, innovation, and digital transformation in Africa	

Key achievements:

- 1. AI Training & Capacity Building Programs: Designed and delivered AI literacy and prompt engineering workshops.
- 2. Designed and delivered training programs for FAIDA (Foundations of AI & Data Analytics (Cohorts 3–4)) and AI Research Clinics (Cohorts 1–4).
- 3. Facilitated Practical AI Tools for Ministry: A deep dive into selected AI tools like NotebookLM for pastoral work in teaching and preaching, research, running events, and content creation.

Key Impact:

- 1. Improved learner confidence and practical use of AI tools
- 2. Enhanced research efficiency through AI-assisted workflows
- 3. Contributed to building an AI-literate learning ecosystem

- i. Identified and evaluated emerging AI-driven product opportunities, translating user needs and market gaps into 65+ actionable innovation concepts for mobile and smart device ecosystems.
- ii. Bridged research and product development by converting conceptual ideas into structured feature proposals, including use cases, functional requirements, and user experience flows.
- iii. Collaborated with cross-functional teams (product managers, engineers, UX designers) to align innovation concepts with technical feasibility and business priorities.
- iv. Conducted localized innovation research (e.g., Swahili language AI use cases), ensuring solutions were contextually relevant for African markets and end users.
- v. Supported product integration strategy by proposing AI features adaptable across devices (smartphones, wearables, apps), enhancing ecosystem synergy.
- vi. Documented and communicated innovation concepts through structured briefs, presentations, and internal proposals to facilitate stakeholder buy-in and potential implementation.
- vii. Contributed to knowledge transfer by sharing insights on AI trends, user behavior, and emerging technologies to inform product roadmaps and innovation pipelines.
- viii. Conducted in-depth interviews and usability tests to understand how African mobile users interact with AI-driven smartphone features.
- ix. Designed and executed multi-modal data collection pipelines to train and evaluate NLP, LLM, and speech recognition models, optimized for mobile devices.
- x. Identified localized user pain points related to data costs, low connectivity, and first-time smartphone usage, informing inclusive AI feature design.
- xi. Translated qualitative and quantitative research findings into user personas, journey maps, and actionable recommendations for engineers and product teams.

INNOVATION & PRODUCT CONTRIBUTIONS – Transsion Holdings

Selected innovations (6 of 65+ concepts across AI, mobile UX, and smart ecosystems)

1. **AI Kids Mode (Facial Recognition Switching)** - Conceptualized an AI-powered feature that detects child users via real-time facial recognition and automatically activates a secure, controlled Kids Mode environment.
2. **Swahili & Hybrid Language Call Summarizer** - Conceptualized a voice AI feature that transcribes and generates concise summaries of calls in Swahili and mixed English-Swahili using speech recognition and NLP.
3. **Voice-Activated Weather Alerts (Smartwatch Integration)** - Proposed a voice-driven weather intelligence system for smartwatches, enabling real-time alerts and hands-free weather queries for proactive user awareness.
4. **Ella Voice-to-Text Engine (Multilingual + Offline)** - Spearheaded integration of a real-time voice-to-text capability within Ella, supporting SMS, WhatsApp, and email with multilingual (Swahili/English) and offline command functionality.
5. **Cross-App Voice Typing Integration (Ella AI Extension)** - Designed API-level extension enabling microphone-based voice typing across third-party apps (e.g., messaging and note-taking apps), improving accessibility and productivity.
6. **AI Message Classifier & Smart Search** - Developed concept for on-device AI that auto-categorizes, tags, and prioritizes messages, enabling faster retrieval through intelligent search.

Key achievements:

1. Led AI usability testing for **voice assistants**, ASR, and NLP-based features, contributing to a 30% improvement in voice command accuracy and driving higher user adoption across African markets.
2. Conducted **generative AI camera feature** research (image enhancement, smart editing, content creation), producing insights that informed product design and enhanced user engagement with AI-powered photo tools.
3. **Translation AI Tool**: Researched and tested AI-driven translation models tailored for African languages, improving accessibility and inclusivity in mobile devices.

Cloud & DevOps Engineer
Kingsway Tyres Ltd

01/2022 to 12/2022
Nairobi, Kenya

- i. Collaborated with development teams to integrate DevOps practices into workflows.
- ii. Monitored system health and resolved issues in real-time environments.
- iii. Documented processes and procedures to improve team knowledge sharing.
- iv. Deployed CI/CD pipelines with Jenkins and Docker to automate the build process.
- v. Managed infrastructure automation using tools like Terraform and Ansible.
- vi. Managed infrastructure, including virtual machines, storage devices, and networks.
- vii. Implemented systems automation using scripting languages like Python and Bash.
- viii. Deployed and supported infrastructure automation and configuration management tools.
- ix. Devised automation, backup, and recovery protocols to preserve and safeguard data.

Key Achievements:

*Strengthened organizational security posture by drafting and implementing comprehensive **IT security policies via Sophos XG240**, reducing unauthorized network access and potential vulnerabilities by 20% through rigorous firewall rule optimization, and SSL inspection.*

Systems & Networks Administrator
Riley Falcon Security Services Ltd

01/2018 to 12/2021
Nairobi, Kenya

- i. Successfully bridged the gap between departmental ICT workflows and business objectives, resulting in an 80% improvement in system uptime and procedural efficiency.
- ii. Provided advanced support for Microsoft 365 products, ensuring a seamless user experience.
- iii. Established and managed security protocols, effectively protecting critical data assets.
- iv. Aligned business processes with Microsoft Dynamics 365 to enhance organizational efficiency.
- v. Deployed Azure resources strategically to maximize service reliability and performance.
- vi. Trained IT personnel on best practices in M365, driving productivity gains.
- vii. Championed the ERP solution implementation on Azure, achieving an 80% operational efficiency boost.
- viii. Facilitated the transition to a paperless environment through M365 integration.

Key Achievements:

1. *Orchestrated the **2019–2021 ICT Strategic Roadmap** for Riley Falcon, aligning multi-year technology initiatives with overarching corporate goals.*
2. *Spearheaded the transition to **ISO 9001:2015-compliant Standard Operating Procedures (SOPs)**, and authored the **2018 ICT Policy** Framework, ensuring 90% operational adherence and streamlined audit readiness.*

Education and Training

Bachelor of Science: Information Technology
KCA University

01/2022
Nairobi

Associate of Applied Science: Aeronautical Engineering (Avionics)
Kenya Aeronautical College

11/2012
Nairobi

Certifications

Teaching the AI Fluency Framework, Anthropic Academy, 2026
AI Fluency for educators, Anthropic Academy, 2026
AI Fluency: Framework & Foundations, Anthropic Academy, 2026
LLMs & LangChain and Workflow Design, Simplilearn, 2026
n8n Workflow Automation, Great Learning, 2026
AI Agent Frameworks, Simplilearn, 2026
n8n Course: No Code AI Agent Builder, Simplilearn, 2026
GitHub Copilot, Microsoft Learn, 2025
Data Governance Fundamentals, Databricks Academy, 2025

AI Agents Fundamentals, Databricks Academy, 2025
OCI AI Foundations Associate, Oracle University, 2025
Ethical AI in Practice, Atingi Online Platform, 2025
Python Libraries for Data Science, Simplilearn, 2025
Managing and Evaluating AI Projects, Atingi Online Platform, 2025
Microsoft Azure AI Essentials, LinkedIn Learning, 2025
DevOps Industrial Training (AWS), IBT Learning, 2025
Data Science & Analytics, HP Foundation, 2024
Artificial Intelligence Fundamentals, IBM, 2024
Big Data and AI, Great Learning Academy, 2024
MLOps Fundamentals, ML Deployment, Great Learning Academy, 2024
Fundamentals of Machine Learning & AI, AWS Training, 2024
STEM Educator: AI Computer Vision for Robotics, Afribots Robotics, 2024
Certified Associate in Project Management (CAPM), Simplilearn, 2024
Certification of Capability in Business Analysis (CCBA), Simplilearn, 2023.

References

- 1. Dr. Japheth Mursi, Data Science Lecturer, Daystar University, jmursi@daystar.ac.ke, +254710469003
- 2. Kevin Kabi, UXR manager, Transsion (Carlcare), kabikevin2@gmail.com, +254728577356
- 3. David Kanjuru, Cloud Support Engineer, AWS, kanjurus8@gmail.com, +254 721 694 125

Languages

English:	C2	French:	A1
Proficient (C2)		Beginner	